

Solution to MP352 Problem Set 4

(1)

P.1

This problem tests a few things: what the terms spacelike, timelike and lightlike mean, how 4-vectors transform if they are written with upper or lower indices, and the invariance of dot products between 4-vectors...

(a) The invariant length-squared of a 4-vector with upper indices

$$\|A\|^2 = -(A^0)^2 + (A^1)^2 + (A^2)^2 + (A^3)^2$$

and if the indices are subscripts

$$\|B\|^2 = -(B_0)^2 + (B_1)^2 + (B_2)^2 + (B_3)^2$$

A^μ is the former k.d., so its norm-squared is

$$\|A\|^2 = -2^2 + 1^2 + (-1)^2 + 0^2 = -4 + 1 + 1 = -2$$

A^μ is a 4-vector whose norm-squared is negative is timelike, so that's what

A^μ is. For B_μ , we have

$$\|B\|^2 = -(-4)^2 + q^2 + 1^2 + 2^2 = -16 + q^2 + 1 + 4 = q^2 - 11$$

Thus, we see $\|B\|^2$ is timelike if $|q| < \sqrt{11}$. If $|q| > \sqrt{11}$, $\|B\|^2 > 0$,

so $|q| > \sqrt{11}$ makes B_μ spacelike. $\|B\|^2 = 0$ is the condition for lightlike,

so $q = \pm \sqrt{11}$ makes B_μ lightlike.

(b) A 4-vector with upper indices will transform as

$$A'^0 = \gamma(v)(A^0 - \frac{v}{c}A^1), A'^1 = \gamma(v)(A^1 - \frac{v}{c}A^0), A'^2 = A^2, A'^3 = A^3$$

if S' moves with velocity $v\hat{e}_x$ relative to S . Thus, in S' , A^μ has components

$$A'^0 = \gamma(v)(2 - \frac{v}{c}), A'^1 = \gamma(v)(1 - 2\frac{v}{c}), A'^2 = -1, A'^3 = 0$$

B_μ , however, transforms with v replaced by $-v$,

$$B'_0 = \gamma(v)(B_0 + \frac{v}{c}B_1), B'_1 = \gamma(v)(B_1 + \frac{v}{c}B_0), B'_2 = B_2, B'_3 = B_3$$

so, for the given B_μ ,

$$B'_0 = \gamma(v)(-4 + \frac{q}{c}), B'_1 = \gamma(v)(q - \frac{4v}{c}), B'_2 = 1, B'_3 = 2$$

(c) $\|A'\|^2 = -(A'^0)^2 + (A'^1)^2 + (A'^2)^2 + (A'^3)^2$, so

$$\|A'\|^2 = -\gamma^2(v)(2 - \frac{v}{c})^2 + \gamma^2(v)(1 - 2\frac{v}{c})^2 + (-1)^2 + 0^2$$

$$= \gamma^2(v) \left[-4 + 4\frac{v}{c} - \frac{v^2}{c^2} + 1 - 4\frac{v}{c} + \frac{4v^2}{c^2} \right] + 1$$

$$= \frac{1}{1 - \frac{v^2}{c^2}} \left[-3 + 3\frac{v^2}{c^2} \right] + 1 = \frac{-3}{1 - \frac{v^2}{c^2}} (1 - \frac{v^2}{c^2}) + 1 = -2$$

which is what we set for $\|A\|^2$ in (a), so $\|A'\|^2 = \|A\|^2$

(2)

$$\|B'\|^2$$

$$\begin{aligned}\|B'\|^2 &= -\gamma^2(v) \left(-4 + \frac{q^2 v^2}{c^2}\right)^2 + \gamma^2(v) \left(q - \frac{4v}{c}\right)^2 + 1^2 + 2^2 \\ &= \gamma^2(v) \left[-16 + 8\frac{q^2 v}{c} - \frac{q^2 v^2}{c^2} + q^2 - 8\frac{q v}{c} + \frac{16v^2}{c^2}\right] + 5 \\ &= \frac{1}{1-\frac{v^2}{c^2}} (q^2 - 16) \left(1 - \frac{v^2}{c^2}\right) + 5 = q^2 - 11\end{aligned}$$

which is $\|B\|^2$, so $\boxed{\|B'\|^2 = \|B\|^2}$

Now, the 4-dot product between the unprimed 4-vector,

$$\begin{aligned}A \cdot B &= A^\mu B_\mu = A^0 B_0 + A^1 B_1 + A^2 B_2 + A^3 B_3 \\ &= 2 \cdot (-4) + 1 \cdot q + (-1) \cdot 1 + 0 \cdot 2 = q - 9\end{aligned}$$

Between primed 4-vector, we have

$$\begin{aligned}A' \cdot B' &= A'^\mu B'_\mu = A'^0 B'_0 + A'^1 B'_1 + A'^2 B'_2 + A'^3 B'_3 \\ &= \gamma^2(v) \left(2 - \frac{4v}{c}\right) \left(-4 + \frac{q^2 v^2}{c^2}\right) + \gamma^2(v) \left(1 - \frac{2v}{c}\right) \left(q - \frac{4v}{c}\right) + (-1)(1) + 0 \cdot 2 \\ &= \gamma^2(v) \left[-8 + (2q + 4)\frac{v}{c} - \frac{q^2 v^2}{c^2} + q - (2q + 4)\frac{v}{c} + \frac{8v^2}{c^2}\right] - 1 \\ &= \frac{1}{1-\frac{v^2}{c^2}} (q - 8) \left(1 - \frac{v^2}{c^2}\right) - 1 = q - 9\end{aligned}$$

so $\boxed{A' \cdot B' = A \cdot B}$ as expected.

P.2

This is stuff you'll see more of if you take MP465 at some point - indeed, the relativistic formulation of electrodynamics takes up the last several weeks of the course. In the problem and the next, we show a bit of what goes on in that formulation...

(a) Note from the def'n that $F_{\mu\nu} = -F_{\nu\mu}$, i.e. it is antisymmetric under interchange of its indices. This means that $F_{00} = F_{11} = F_{22} = F_{33} = 0$ and that once we know $F_{01}, F_{02}, F_{03}, F_{12}, F_{23}$ and F_{31} , we know $F_{10}, F_{20}, F_{30}, F_{21}, F_{32}$ and F_{13} as well - the only six independent components. These are, in fact, the six components of the two vectors $\vec{E}$ and $\vec{B}$, as we now see:

$$F_{01} = \partial_0 A_1 - \partial_1 A_0; \quad \partial_0 = \frac{\partial}{\partial x^0} = \frac{\partial}{\partial(ct)} = \frac{1}{c} \frac{\partial}{\partial t} \quad \text{and} \quad \partial_1 = \frac{\partial}{\partial x^1} = \frac{\partial}{\partial x}.$$

$A_1 = A^1 = A_x$ (recall that spatial indices can be raised and lowered with no change in sign) and $A_0 = -A^0 = -\Phi/c$ (the sign changes when we lower or raise a time index), so

$$F_{01} = \frac{1}{c} \frac{\partial}{\partial t} (A_x) - \frac{\partial}{\partial x} \left(-\frac{\Phi}{c}\right) = \frac{1}{c} \frac{\partial \Phi}{\partial x} + \frac{1}{c} \frac{\partial A_x}{\partial t}$$

But $\vec{E} = -\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t}$ means $E_x = -\frac{\partial \Phi}{\partial x} - \frac{\partial A_x}{\partial t}$, so this means that

$$\boxed{F_{01} = -E_x/c}$$

②

$$F_{02} = \partial_0 A_2 - \partial_2 A_0 : \text{this is very similar, since } \partial_2 = \frac{\partial}{\partial x_2} = \frac{\partial}{\partial y} \text{ and } A_2 = A^2 = A_y,$$

giving

$$F_{02} = \frac{1}{c} \frac{\partial}{\partial t} (A_y) - \frac{\partial}{\partial y} (-\Phi/c) = \frac{1}{c} \frac{\partial \Phi}{\partial y} + \frac{1}{c} \frac{\partial A_y}{\partial t} = -\frac{1}{c} \left(-\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t} \right)_y$$

$$\boxed{E_y / c}$$

$$F_{03} = \partial_0 A_3 - \partial_3 A_0 : A_3 = A^3 = A_z \text{ and } \partial_3 = \frac{\partial}{\partial x_3} = \frac{\partial}{\partial z}, \text{ so}$$

$$F_{03} = \frac{1}{c} \frac{\partial}{\partial t} (A_z) - \frac{\partial}{\partial z} (-\Phi/c) = \frac{1}{c} \frac{\partial \Phi}{\partial z} + \frac{1}{c} \frac{\partial A_z}{\partial t} = -\frac{1}{c} \left(-\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t} \right)_z$$

$$\boxed{-E_z / c}$$

$$F_{12} = \partial_1 A_2 - \partial_2 A_1 : \text{from what we've done above, we see } F_{12} = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}. \text{ But}$$

$$\text{note that } \vec{\nabla} \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{e}_x + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{e}_y + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{e}_z$$

$$\text{so } F_{12} = (\vec{\nabla} \times \vec{A})_z. \text{ But } \vec{B} = \vec{\nabla} \times \vec{A} \text{ is the magnetic field, so}$$

$$\boxed{F_{12} = B_z}$$

$$F_{23} = \partial_2 A_3 - \partial_3 A_2 : \text{this is } \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}, \text{ which is } (\vec{\nabla} \times \vec{A})_x. \text{ Thus,}$$

$$\boxed{F_{23} = B_x}$$

$$F_{31} = \partial_3 A_1 - \partial_1 A_3 : \text{this is } \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} = (\vec{\nabla} \times \vec{A})_y, \text{ so } \boxed{F_{31} = B_y}$$

(Exercise for the interested: use these results to show that the two

Maxwell eq's $\vec{\nabla} \cdot \vec{B} = 0$ and $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ can be written as

$$\partial_\mu F^{\mu\nu} = 0.)$$

$$(b) \text{ Let's write this out in full: } \frac{1}{c} \frac{\partial \Phi}{\partial t} + \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 0. \text{ Now if}$$

$$\text{we use } \frac{\partial}{\partial t} = c \partial_0, \frac{\partial}{\partial x} = \partial_1, \frac{\partial}{\partial y} = \partial_2, \frac{\partial}{\partial z} = \partial_3, \Phi = c A^0, A_x = A^1,$$

$$A_y = A^2 \text{ and } A_z = A^3, \text{ we get}$$

$$\frac{1}{c} (c \partial_0)(c A^0) + (\partial_1)(A^1) + (\partial_2)(A^2) + (\partial_3)(A^3)$$

$$= \partial_0 A^0 + \partial_1 A^1 + \partial_2 A^2 + \partial_3 A^3 = 0$$

so we have a lower and upper index summed over all possible values, from 0 to 3, this can therefore be written in Einstein's summation convention as,

$$\boxed{\partial_\mu A^\mu = 0}$$

This is called the "Lorenz gauge condition", and its important is that it's relativistically invariant. Beac ∂_μ and A^μ transform as

4-vectors, if we contract their indices, we get a Lorentz invariant, i.e.

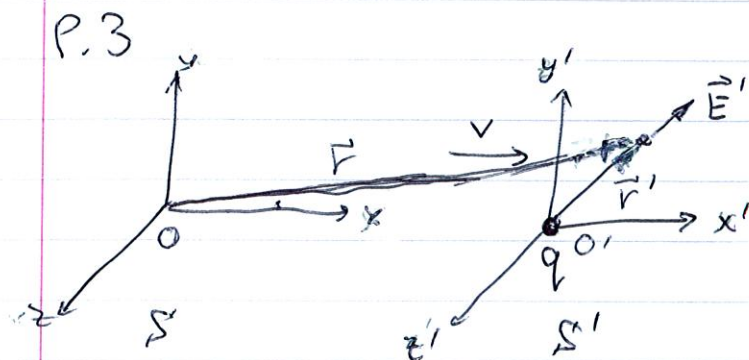
the value of $\partial_\mu A^\mu$ is same in all frames S as the value of $\partial'_\mu A'^\mu$ in

a different inertial frame S' . Thus, if $\partial_\mu A^\mu$ is zero in any frame, it's

zero in all frames.

(4)

(Trivia: despite the similarity in their names, the Lorenz - not "t" - whom the Lorenz gauge condition $\partial_\mu A^\mu = 0$ is named after is not the same Lorentz - with a "t" - who gave us his transformations. Since they were contemporaries of each other, I wonder if they ever talked about this whenever they met: "Hey, those are MY transformations, not yours!" "Oh yeah? Give me back my gauge condition, you schweinshund!")



The drawing above illustrates the situation: in S' , q is at rest and so the electric potential and field are just those of a point charge at the origin O' , namely, $\Phi' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{x'^2 + y'^2 + z'^2}}$ and $\vec{A}' = 0$ (since there is no magnetic field).

(a) Since A^μ is a 4-vector, we know exactly how to relate its components in S' to the ones given above in S' :

$$A^0 = \gamma(v) \left(A'^0 + \frac{v}{c} A'^1 \right) \Rightarrow \Phi/c = \gamma(v) \Phi'/c$$

$$A^1 = \gamma(v) \left(A'^1 + \frac{v}{c} A'^0 \right) \Rightarrow A_x = \gamma(v) v/c \Phi'/c$$

$$A^2 = A'^2 \Rightarrow A_y = 0$$

$$A^3 = A'^3 \Rightarrow A_z = 0$$

so we get

$$\Phi = \gamma(v) \Phi' = \frac{\gamma(v) q}{4\pi\epsilon_0 \sqrt{(x')^2 + (y')^2 + (z')^2}}$$

$$= \frac{\gamma(v) q}{4\pi\epsilon_0 \sqrt{\gamma^2(v) (x-vt)^2 + y^2 + z^2}}$$

$$\vec{A} = \gamma(v) v/c^2 \Phi' \hat{e}_x = \left[\frac{1}{4\pi\epsilon_0} \frac{\gamma(v) v q / c^2}{\sqrt{\gamma^2(v) (x-vt)^2 + y^2 + z^2}} \right] \hat{e}_x$$

where we must use $x' = \gamma(v)(x-vt)$, $y' = y$ and $z' = z$, since we want all quantities in terms of what we can measure in S , not S' . We can rewrite $\vec{A}$ slightly using the identity $c^2 = \frac{1}{\mu_0 \epsilon_0}$ to get the more magnetic-looking

$$\vec{A} = \frac{\mu_0}{4\pi c} \frac{\gamma(v) v q}{\sqrt{\gamma^2(v)(x-vt)^2 + y^2 + z^2}} \hat{e}_x \quad (5)$$

(b) Now that we have Φ & $\vec{A}$ we can get $\vec{E}$ & $\vec{B}$:

$$\vec{E} = -\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t}$$

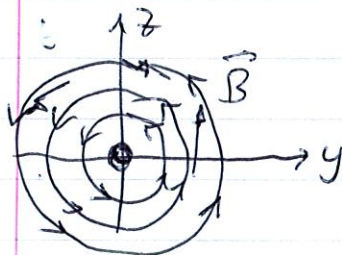
$$\begin{aligned} &= -\vec{\nabla} \left[\frac{1}{4\pi\epsilon_0} \frac{\gamma(v)q}{\sqrt{\gamma^2(v)(x-vt)^2 + y^2 + z^2}} \right] - \frac{\partial}{\partial t} \left[\frac{1}{4\pi\epsilon_0} \frac{\gamma(v)vq/c^2}{\sqrt{\gamma^2(v)(x-vt)^2 + y^2 + z^2}} \hat{e}_x \right] \\ &= \frac{1}{4\pi\epsilon_0} \left\{ \frac{\gamma^3(v)(x-vt)q - \gamma^3(v)v^2/c^2(x-vt)q}{[\gamma^2(v)(x-vt)^2 + y^2 + z^2]^{3/2}} \hat{e}_x \right. \\ &\quad \left. + \frac{\gamma(v)q y \hat{e}_y + \gamma(v)q z \hat{e}_z}{[\gamma^2(v)(x-vt)^2 + y^2 + z^2]^{3/2}} \right\} \\ &= \boxed{\frac{1}{4\pi\epsilon_0} \frac{\gamma(v)q [(x-vt)\hat{e}_x + y\hat{e}_y + z\hat{e}_z]}{[\gamma^2(v)(x-vt)^2 + y^2 + z^2]^{3/2}}} \end{aligned}$$

The magnetic field is

$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{\partial A_x}{\partial z} \hat{e}_y - \frac{\partial A_x}{\partial y} \hat{e}_z \quad (\text{since } \vec{A} \text{ only has an } x\text{-component})$$

$$\begin{aligned} &= \frac{\mu_0}{4\pi} \left\{ \frac{\partial}{\partial z} \left(\frac{\gamma(v)vq}{\sqrt{\gamma^2(v)(x-vt)^2 + y^2 + z^2}} \right) \hat{e}_y - \frac{\partial}{\partial y} \left(\frac{\gamma(v)vq}{\sqrt{\gamma^2(v)(x-vt)^2 + y^2 + z^2}} \right) \hat{e}_z \right\} \\ &= \boxed{\frac{\mu_0}{4\pi} \frac{\gamma(v)vq (-z\hat{e}_y + y\hat{e}_z)}{[\gamma^2(v)(x-vt)^2 + y^2 + z^2]^{3/2}}} \end{aligned}$$

Note that $\vec{E}$ points radially out from the charge, not just out at



$\vec{B}$ points around the direction of motion: if the charge is coming out of the page like in the diagram to the left, $\vec{B}$ "wraps around" the line of motion. This is exactly the direction of $\vec{B}$ due to a current element, but that's no surprise:

a current element is just a bunch of point charges moving with a constant velocity, so it's expected that $\vec{B}$ will have the same behaviour. (In fact, if we define

(c) OPTIONAL: do these satisfy Maxwell's eq's? Well, for $\vec{r} \neq (x-vt)\hat{e}_x + y\hat{e}_y + z\hat{e}_z$ (where the charge is), do $\vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot \vec{B} = 0$ and $\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = \vec{\nabla} \times \vec{B} - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \vec{0}$ hold?

(7)

Yes, try do: so $\vec{B} = \vec{\nabla} \times \vec{A}$, $\vec{\nabla} \cdot \vec{B} = 0$ is automatic (the divergence of a curl is always zero). And via $\vec{\nabla} \times \vec{E} = \vec{\nabla} \times (-\vec{\nabla} \Phi - \frac{\partial \vec{A}}{\partial t})$
 $= -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{A}) = -\frac{\partial \vec{B}}{\partial t}$ (the curl of a gradient is always zero), this
 is automatic as well. Thus, we need only check $\vec{\nabla} \cdot \vec{E} = 0$ & $\vec{\nabla} \times \vec{B} =$
 $\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \vec{0}$.

$$\vec{\nabla} \cdot \vec{E} = \frac{\gamma(t)q}{4\pi\epsilon_0} \left\{ \frac{\partial}{\partial x} \left[\frac{x-vt}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right] + \frac{\partial}{\partial y} \left[\frac{y}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right] + \frac{\partial}{\partial z} \left[\frac{z}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right] \right\}$$

$$= \frac{\gamma(t)q}{4\pi\epsilon_0} \left\{ \frac{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2} - 2\gamma^2(t)(x-vt)^2 + [\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2} + [\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2} - 2yz^2}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{5/2}} \right\}$$

$$= 0$$

$$\vec{\nabla} \times \vec{B} = \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \hat{e}_x - \frac{\partial B_z}{\partial x} \hat{e}_y + \frac{\partial B_y}{\partial x} \hat{e}_z \quad (\text{since } B_x = 0)$$

$$= \frac{\mu_0 \gamma(t) v q}{4\pi} \left\{ \left[\frac{\partial}{\partial y} \left[\frac{y}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right] - \frac{\partial}{\partial z} \left[\frac{-z}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right] \right] \hat{e}_x \right.$$

$$\left. - \frac{\partial}{\partial x} \left[\frac{y}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right] \hat{e}_y + \frac{\partial}{\partial x} \left[\frac{-z}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right] \hat{e}_z \right\}$$

$$= \frac{\mu_0 \gamma(t) v q}{4\pi} \left\{ \frac{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2} - 2yz^2 + [\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{5/2}} \hat{e}_x \right.$$

$$\left. + \frac{\gamma^2(t)(x-vt)(y\hat{e}_y + z\hat{e}_z)}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right\}$$

$$= \frac{\mu_0 \gamma(t) v q}{4\pi} \frac{[2\gamma^2(t)(x-vt)^2+y^2+z^2] \hat{e}_x + \gamma^2(t)(x-vt)(y\hat{e}_y + z\hat{e}_z)}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}}$$

$$\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \epsilon_0 \frac{\gamma(t)q}{4\pi\epsilon_0} \frac{\partial}{\partial t} \left[\frac{(x-vt)\hat{e}_x + y\hat{e}_y + z\hat{e}_z}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right]$$

$$= \frac{\mu_0 \gamma(t) q}{4\pi} \left\{ \frac{-v[y^2+z^2-2\gamma^2(t)(x-vt)^2] \hat{e}_x + v\gamma^2(t)(x-vt)(y\hat{e}_y + z\hat{e}_z)}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{5/2}} \right\}$$

$$= \frac{\mu_0 \gamma(t) v q}{4\pi} \left\{ \frac{[2\gamma^2(t)(x-vt)^2+y^2+z^2] \hat{e}_x + \gamma^2(t)(x-vt)(y\hat{e}_y + z\hat{e}_z)}{[\gamma^2(t)(x-vt)^2+y^2+z^2]^{3/2}} \right\}$$

$$= \boxed{\vec{\nabla} \times \vec{B}}$$

so Maxwell's eq's are indeed satisfied, as they must be.
 Yay.